Corey Oses
Materials Science and Engineering, Johns Hopkins University
Physics and Astronomy (joint), Johns Hopkins University

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Work Experience

Assistant Professor — 2022–present — Johns Hopkins University
- Joint appointment, Department of Physics and Astronomy (2025–present)

Postdoctoral Fellow — 2018–2022 — Duke University

Education

Ph.D. — 2013–2018 — Duke University

Department: Mechanical Engineering and Materials Science
Thesis: *Machine learning, phase stability, and disorder with the Automatic Flow Framework for Materials Discovery*
DukeSpace: hdl.handle.net/10161/18254

B.Sc. — 2009–2013 — Cornell University

Department: Applied and Engineering Physics
Thesis: *Plume Propagation Simulation for Pulsed Laser Deposition*

Honors and Awards

Award	2026	Finalist, Advanced Nuclear Energy Technology Challenge, TechConnect
Award	2026	JHU Catalyst Early Career Award, Johns Hopkins University
Award	2026	MGB-SIAM Early Career Fellowship, Society for Industrial and Applied Mathematics
Award	2026	Early Career Award: Inspiring Generations of New Innovators to Impact Technologies in Energy 2025 (IGNIITE 2025), Advanced Research Projects Agency-Energy (ARPA-E)
Publication Award	2026	Most Popular Articles Collection, Publication in Mater. Horiz., Royal Society of Chemistry: Materials Horizons
Award	2024	Early Career Investigator Award in Materials Modeling, International Society of Materials Modeling
Publication Award	2024	Editors' Highlight, Publication in Nat. Commun., Springer Nature
Award	2022	Reviewer of the Year, npj Computational Materials
Publication Award	2022	Editor's Choice, Publication in Comput. Mater. Sci., Elsevier

Publication Award	Nov 16, 2021	"Hot paper", Publication in Nat. Rev. Mater., Web of Science (Clarivate Analytics) • Published in the past two years and received enough citations in July/August 2021 to place it in the top 0.1% of papers in the academic field of Materials Science
Publication Award	2021	Editors' Highlight, Publication in Nat. Commun., Springer Nature
Publication Award	2018	Editor's Choice, Publication in Comput. Mater. Sci., Elsevier
Publication Award	2017	Editor's Choice, Publication in Comput. Mater. Sci., Elsevier
Award	Aug 14, 2015	Best Teaching Assistant Award (ME 221), Duke University Department of Mechanical Engineering and Materials Science
Publication Award	2015	Editor's Choice, Publication in Comput. Mater. Sci., Elsevier
Publication Award	2015	Top 10 most highly downloaded papers for the month of January 2015, Publication in Chem. Mater., American Chemical Society
Publication Award	2015	Editors' Choice, Publication in Chem. Mater., American Chemical Society
Fellowship	2013–2016	Graduate Research Fellowship, National Science Foundation

Journal Publications † contributed equally * corresponding

2026

50. S.-Y. Tseng, G. Han, T. Li, X. Xu, J. Hu, G. Qiu & C. Oses*, *Structural benchmarks overlook density-of-states errors in machine-learned interatomic potentials*, APL Mach. Learn. **in press** (2026). **DOI**: 10.1063/5.0339982.
49. X. Xu, G. Han, T. Li, J. Hu, G. Qiu, S.-Y. Tseng, G. H. Han, A. Dupuy, A. S. Hall & C. Oses*, *Stabilizing Iodine in Pyrochlore: Toward New Nuclear Waste Forms*, Precis. Chem. **4**(8), 1211–1224 (2026). **DOI**: 10.1021/prechem.5c00467. [PDF]

2025

48. G. Han†, T. Li†, X. Xu, J. Lee, S. Sequeira, A. Ajith & C. Oses*, *The search for high-entropy fuel-cell catalysts using disorder descriptors*, Nano Futures **9**, 045001 (2025). **DOI**: 10.1088/2399-1984/ae19b0. [PDF]
47. C. Oses*, T. Li, X. Xu, G. Han, G. Qiu & J. R. Owens, *Beyond the four core effects: revisiting thermoelectrics with a high-entropy design*, Mater. Horiz. **12**, 5946–5956 (2025). **DOI**: 10.1039/D5MH00356C. [PDF]
    - This paper was selected to feature in 2025 Most Popular Articles by the Royal Society of Chemistry: Materials Horizons (2026).
46. G. Qiu, T. Li, X. Xu, Y. Liu, M. Niyogi, K. Cariaga & C. Oses*, *High entropy powering green energy: hydrogen, batteries, electronics, and catalysis*, npj Comput. Mater. **11**, 145 (2025). **DOI**: 10.1038/s41524-025-01594-6. [PDF]

2024

45. T. Gong, G. Qiu, M.-R. He, O. V. Safonova, W.-C. Yang, D. Raciti, C. Oses & A. S. Hall, *Atomic Ordering-Induced Ensemble Variation in Alloys Governs Electrocatalyst On/Off States*, J. Am. Chem. Soc. **147**(1), 510–518 (2024). **DOI**: 10.1021/jacs.4c11753.
44. K. S. Vecchio, S. Curtarolo, K. Kaufmann, T. J. Harrington, C. Oses & C. Toher, *Fermi energy engineering of enhanced plasticity in high-entropy carbides*, Acta Mater. **276**, 120117 (2024). **DOI**: 10.1016/j.actamat.2024.120117. [PDF]
43. M. L. Evans, J. Bergsma, A. Merkys, C. W. Andersen, O. B. Andersson, D. Beltrán, E. Blokhin, T. M. Boland, R. Castañeda Balderas, K. Choudhary, A. Díaz Díaz, R. Domínguez García, H. Eckert, K. Eimre, M. E. Fuentes-Montero, A. M. Krajewski, J. J. Mortensen, J. M. Nápoles-Duarte, J. Pietryga, J. Qi, F. d. J. Trejo Carrillo, A. Vaitkus, J. Yu, A. C. Zettel, P. Baptista de Castro, J. Carlsson, T. F. T. Cerqueira, S. Divilov, H. Hajiyani, F. Hanke, K. Jose, C. Oses, J. Riebesell, J. Schmidt, D. Winston, C. Xie, X. Yang, S. Bonella, S. Botti, S. Curtarolo, C. Draxl, L. E. Fuentes Cobas, A. Hospital, Z.-K. Liu, M. A. L. Marques, N. Marzari, A. J. Morris, S. P. Ong, M. Orozco, K. A. Persson, K. S. Thygesen, C. Wolverton, M. Scheidgen, C. Toher, G. J. Conduit, G. Pizzi, S. Gražulis, G.-M. Rignanese & R. Armiento, *Developments and applications of the OPTIMADE API for materials discovery, design, and data exchange*, Digit. Discov. **3**, 1509–1533 (2024). **DOI**: 10.1039/D4DD00039K. [PDF]
42. S. Divilov†, H. Eckert†, D. Hicks†, C. Oses†, C. Toher†, R. Friedrich, M. Esters, M. J. Mehl, A. C. Zettel, Y. Lederer, E. Zurek, J.-P. Maria, D. W. Brenner, X. Campilongo, S. Filipović, W. G. Fahrenholtz, C. J. Ryan, C. M. DeSalle, R. J. Crealese, D. E. Wolfe, A. Calzolari & S. Curtarolo, *Disordered enthalpy-entropy descriptor for high-entropy ceramics discovery*, Nature **625**, 66–73 (2024). **DOI**: 10.1038/s41586-023-06786-y. [PDF]

41. A. B. Peters, D. Zhang, S. Chen, C. Ott, C. Oses, S. Curtarolo, I. McCue, T. Pollock & S. E. Prameela, *Materials Design for Hypersonics*, Nat. Commun. **15**, 3328 (2024). **DOI**: 10.1038/s41467-024-46753-3. [PDF]
    - This paper was selected for Editors' Highlight by Springer Nature (2024).

2023

40. D. E. Wolfe, C. M. DeSalle, C. J. Ryan, R. E. Slapikas, R. T. Sweny, R. J. Crealese, P. A. Kolonin, S. P. Stepanoff, A. Haque, S. Divilov, H. Eckert, C. Oses, M. Esters, D. W. Brenner, W. G. Fahrenholtz, J.-P. Maria, C. Toher, E. Zurek & S. Curtarolo, *Influence of Processing on the Microstructural Evolution and Multiscale Hardness in Titanium Carbonitrides (TiCN) Produced via Field Assisted Sintering Technology*, Materialia **27**, 101682 (2023). **DOI**: 10.1016/j.mtla.2023.101682.
39. C. Oses, M. Esters, D. Hicks, S. Divilov, H. Eckert, R. Friedrich, M. J. Mehl, A. Smolyanyuk, X. Campilongo, A. van de Walle, J. Schroers, A. G. Kusne, I. Takeuchi, E. Zurek, M. Buongiorno Nardelli, M. Fornari, Y. Lederer, O. Levy, C. Toher & S. Curtarolo, *aflow++: a C++ framework for autonomous materials design*, Comput. Mater. Sci. **217**, 111889 (2023). **DOI**: 10.1016/j.commatsci.2022.111889. [preprint]
    - This paper was selected for Editor's Choice by Elsevier (2022).
38. M. Esters, C. Oses, S. Divilov, H. Eckert, R. Friedrich, D. Hicks, M. J. Mehl, F. Rose, A. Smolyanyuk, A. Calzolari, X. Campilongo, C. Toher & S. Curtarolo, *aflow.org: a web ecosystem of databases, software and tools*, Comput. Mater. Sci. **216**, 111808 (2023). **DOI**: 10.1016/j.commatsci.2022.111808. [preprint]
37. M. Esters†, A. Smolyanyuk†, C. Oses, D. Hicks, S. Divilov, H. Eckert, X. Campilongo, C. Toher & S. Curtarolo, *QH-POCC: taming tiling entropy in thermal expansion calculations of disordered materials*, Acta Mater. **245**, 118594 (2023). **DOI**: 10.1016/j.actamat.2022.118594. [preprint]

2022

36. A. Calzolari, C. Oses, C. Toher, M. Esters, X. Campilongo, S. P. Stepanoff, D. E. Wolfe & S. Curtarolo, *Plasmonic high-entropy carbides*, Nat. Commun. **13**, 5993 (2022). **DOI**: 10.1038/s41467-022-33497-1. [PDF]
35. X. Wang, D. M. Proserpio, C. Oses, C. Toher, S. Curtarolo & E. Zurek, *The Microscopic Diamond Anvil Cell: Stabilization of Superhard, Superconducting Carbon Allotropes at Ambient Pressure*, Angew. Chem. **61**(32), e202205129 (2022). **DOI**: 10.1002/anie.202205129.
34. H. J. Kulik, T. Hammerschmidt, J. Schmidt, S. Botti, M. A. L. Marques, M. Boley, M. Scheffler, M. Todorović, P. Rinke, C. Oses, A. Smolyanyuk, S. Curtarolo, A. Tkatchenko, A. P. Bartók, S. Manzhos, M. Ihara, T. Carrington, J. Behler, O. Isayev, M. Veit, A. Grisafi, J. Nigam, M. Ceriotti, K. T. Schütt, J. Westermayr, M. Gastegger, R. J. Maurer, B. Kalita, K. Burke, R. Nagai, R. Akashi, O. Sugino, J. Hermann, F. Noé, S. Pilati, C. Draxl, M. Kuban, S. Rigamonti, M. Scheidgen, M. Esters, D. Hicks, C. Toher, P. V. Balachandran, I. Tamblyn, S. Whitelam, C. Bellinger & L. M. Ghiringhelli, *Roadmap on Machine Learning in Electronic Structure*, Electron. Struct. **4**(2), 023004 (2022). **DOI**: 10.1088/2516-1075/ac572f. [PDF]
33. A. G. Kusne, A. McDannald, B. DeCost, C. Oses, C. Toher, S. Curtarolo, A. Mehta & I. Takeuchi, *Physics in the Machine: Integrating Physical Knowledge in Autonomous Phase-Mapping*, Front. Phys. **10**, 815863 (2022). **DOI**: 10.3389/fphy.2022.815863. [PDF]
32. C. Toher, C. Oses, M. Esters, D. Hicks, G. N. Kotsonis, C. M. Rost, D. W. Brenner, J.-P. Maria & S. Curtarolo, *High-entropy ceramics: Propelling applications through disorder*, MRS Bull. **47**, 194–202 (2022). **DOI**: 10.1557/s43577-022-00281-x.

2021

31. M. Esters, C. Oses, D. Hicks, M. J. Mehl, M. Jahnátek, M. D. Hossain, J.-P. Maria, D. W. Brenner, C. Toher & S. Curtarolo, *Settling the matter of the role of vibrations in the stability of high-entropy carbides*, Nat. Commun. **12**, 5747 (2021). **DOI**: 10.1038/s41467-021-25979-5. [PDF]
    - This paper was selected for Editors' Highlight by Springer Nature (2021).
30. M. D. Hossain, T. Borman, C. Oses, M. Esters, C. Toher, L. Feng, A. Kumar, W. G. Fahrenholtz, S. Curtarolo, D. W. Brenner, J. M. LeBeau & J.-P. Maria, *Entropy Landscaping of High-Entropy Carbides*, Adv. Mater. **33**(42), 2102904 (2021). **DOI**: 10.1002/adma.202102904.
29. C. W. Andersen†, R. Armiento†, E. Blokhin†, G. J. Conduit†, S. Dwaraknath†, M. L. Evans†, Á. Fekete†, A. Gopakumar†, S. Gražulis†, A. Merkys†, F. Mohamed†, C. Oses†, G. Pizzi†, G.-M. Rignanese†, M. Scheidgen†, L. Talirz†, C. Toher†, D. Winston†, R. Aversa, K. Choudhary, P. Colinet, S. Curtarolo, D. Di Stefano, C. Draxl, S. Er, M. Esters, M. Fornari, M. Giantomassi, M. Govoni, G. Hautier, V. Hegde, M. K. Horton, P. Huck, G. Huhs, J. Hummelshøj, A. Kariryaa, B. Kozinsky, S. Kumbhar, M. Liu, N. Marzari, A. J. Morris, A. Mostofi, K. A. Persson, G. Petretto, T. Purcell, F. Ricci, F. Rose, M. Scheffler, D. Speckhard, M. Uhrin, A. Vaitkus, P. Villars, D. Waroquiers, C. Wolverton, M. Wu & X. Yang, *OPTIMADE: an API for exchanging materials data*, Sci. Data **8**, 217 (2021). **DOI**: 10.1038/s41597-021-00974-z. [PDF]
28. R. Friedrich, M. Esters, C. Oses, S. Ki, M. J. Brenner, D. Hicks, M. J. Mehl, C. Toher & S. Curtarolo, *Automated coordination corrected enthalpies with AFLOW-CCE*, Phys. Rev. Mater. **5**, 043803 (2021). **DOI**: 10.1103/PhysRevMaterials.5.043803.
27. D. Hicks, M. J. Mehl, M. Esters, C. Oses, O. Levy, G. L. W. Hart, C. Toher & S. Curtarolo, *The AFLOW Library of Crystallographic Prototypes: Part 3*, Comput. Mater. Sci. **199**, 110450 (2021). **DOI**: 10.1016/j.commatsci.2021.110450.
26. M. J. Mehl, M. Ronquillo, D. Hicks, M. Esters, C. Oses, R. Friedrich, A. Smolyanyuk, E. Gossett, D. Finkenstadt & S. Curtarolo, *Tin-pest problem as a test of density functionals using high-throughput calculations*, Phys. Rev. Mater. **5**, 083608 (2021). **DOI**: 10.1103/PhysRevMaterials.5.083608.
25. M. D. Hossain†, T. Borman†, A. Kumar, X. Chen, A. Khosravani, S. R. Kalidindi, E. A. Paisley, M. Esters, C. Oses, C. Toher, S. Curtarolo, J. M. LeBeau, D. W. Brenner & J.-P. Maria, *Carbon Stoichiometry and Mechanical Properties of High Entropy Carbides*, Acta Mater. **215**, 117051 (2021). **DOI**: 10.1016/j.actamat.2021.117051.

2020

24. A. G. Kusne†, H. Yu†, C. Wu, H. Zhang, J. Hattrick-Simpers, B. DeCost, S. Sarker, C. Oses, C. Toher, S. Curtarolo, A. V. Davydov, R. Agarwal, L. A. Bendersky, M. Li, A. Mehta & I. Takeuchi, *On-the-fly Closed-loop Autonomous Materials Discovery via Bayesian Active Learning*, Nat. Commun. **11**, 5966 (2020). **DOI**: 10.1038/s41467-020-19597-w. [PDF]
23. K. Kaufmann, D. Maryanovsky, W. M. Mellor, C. Zhu, A. S. Rosengarten, T. J. Harrington, C. Oses, C. Toher, S. Curtarolo & K. S. Vecchio, *Discovery of novel high-entropy ceramics via machine learning*, npj Comput. Mater. **6**, 42 (2020). **DOI**: 10.1038/s41524-020-0317-6. [PDF]
22. C. Oses, C. Toher & S. Curtarolo, *High-entropy ceramics*, Nat. Rev. Mater. **5**, 295–309 (2020). **DOI**: 10.1038/s41578-019-0170-8.
    - This paper was highlighted as a "hot paper" by Web of Science (Clarivate Analytics) (Nov 16, 2021).

2019

21. D. C. Ford, D. Hicks, C. Oses, C. Toher & S. Curtarolo, *Metallic glasses for biodegradable implants*, Acta Mater. **176**, 297–305 (2019). **DOI**: 10.1016/j.actamat.2019.07.008.
20. P. Avery, X. Wang, C. Oses, E. Gossett, D. M. Proserpio, C. Toher, S. Curtarolo & E. Zurek, *Predicting Superhard Materials via a Machine Learning Informed Evolutionary Structure Search*, npj Comput. Mater. **5**, 89 (2019). **DOI**: 10.1038/s41524-019-0226-8. [PDF]
19. C. Toher, C. Oses, D. Hicks & S. Curtarolo, *Unavoidable disorder and entropy in multi-component systems*, npj Comput. Mater. **5**, 69 (2019). **DOI**: 10.1038/s41524-019-0206-z. [PDF]
18. R. Friedrich, D. Usanmaz, C. Oses, A. R. Supka, M. Fornari, M. Buongiorno Nardelli, C. Toher & S. Curtarolo, *Coordination corrected ab initio formation enthalpies*, npj Comput. Mater. **5**, 59 (2019). **DOI**: 10.1038/s41524-019-0192-1. [PDF]
17. P. Nath, D. Usanmaz, D. Hicks, C. Oses, M. Fornari, M. Buongiorno Nardelli, C. Toher & S. Curtarolo, *AFLOW-QHA3P: Robust and automated method to compute thermodynamic properties of solids*, Phys. Rev. Mater. **3**, 073801 (2019). **DOI**: 10.1103/PhysRevMaterials.3.073801.

2018

16. C. Oses, E. Gossett, D. Hicks, F. Rose, M. J. Mehl, E. Perim, I. Takeuchi, S. Sanvito, M. Scheffler, Y. Lederer, O. Levy, C. Toher & S. Curtarolo, *AFLOW-CHULL: Cloud-oriented platform for autonomous phase stability analysis*, J. Chem. Inf. Model. **58**(12), 2477–2490 (2018). **DOI**: 10.1021/acs.jcim.8b00393.
15. C. Oses, C. Toher & S. Curtarolo, *Data-driven design of inorganic materials with the Automatic Flow Framework for Materials Discovery*, MRS Bull. **43**(9), 670–675 (2018). **DOI**: 10.1557/mrs.2018.207.
14. P. Sarker†, T. J. Harrington†, C. Toher, C. Oses, M. Samiee, J.-P. Maria, D. W. Brenner, K. S. Vecchio & S. Curtarolo, *High-entropy high-hardness metal carbides discovered by entropy descriptors*, Nat. Commun. **9**, 4980 (2018). **DOI**: 10.1038/s41467-018-07160-7. [PDF]
13. V. Stanev, C. Oses, A. G. Kusne, E. Rodriguez, J. Paglione, S. Curtarolo & I. Takeuchi, *Machine learning modeling of superconducting critical temperature*, npj Comput. Mater. **4**, 29 (2018). **DOI**: 10.1038/s41524-018-0085-8. [PDF]
12. E. Gossett, C. Toher, C. Oses, O. Isayev, F. Legrain, F. Rose, E. Zurek, J. Carrete, N. Mingo, A. Tropsha & S. Curtarolo, *AFLOW-ML: A RESTful API for machine-learning prediction of materials properties*, Comput. Mater. Sci. **152**, 134–145 (2018). **DOI**: 10.1016/j.commatsci.2018.03.075.
    - This paper was selected for Editor's Choice by Elsevier (2018).
11. D. Hicks, C. Oses, E. Gossett, G. Gomez, R. H. Taylor, C. Toher, M. J. Mehl, O. Levy & S. Curtarolo, *AFLOW-SYM: platform for the complete, automatic and self-consistent symmetry analysis of crystals*, Acta Cryst. A **74**, 184–203 (2018). **DOI**: 10.1107/S2053273318003066.

2017

10. A. Hever, C. Oses, S. Curtarolo, O. Levy & A. Natan, *The structure and composition statistics of 6A binary and ternary structures*, Inorg. Chem. **57**(2), 653–667 (2017). **DOI**: 10.1021/acs.inorgchem.7b02462.
9. F. Rose, C. Toher, E. Gossett, C. Oses, M. Buongiorno Nardelli, M. Fornari & S. Curtarolo, *AFLUX: The LUX materials search API for the AFLOW data repositories*, Comput. Mater. Sci. **137**, 362–370 (2017). **DOI**: 10.1016/j.commatsci.2017.04.036.
    - This paper was selected for Editor's Choice by Elsevier (2017).
8. O. Isayev†, C. Oses†, C. Toher, E. Gossett, S. Curtarolo & A. Tropsha, *Universal Fragment Descriptors for Predicting Properties of Inorganic Crystals*, Nat. Commun. **8**, 15679 (2017). **DOI**: 10.1038/ncomms15679. [PDF]
7. C. Toher, C. Oses, J. J. Plata, D. Hicks, F. Rose, O. Levy, M. de Jong, M. Asta, M. Fornari, M. Buongiorno Nardelli & S. Curtarolo, *Combining the AFLOW GIBBS and elastic libraries to efficiently and robustly screen thermomechanical properties of solids*, Phys. Rev. Mater. **1**, 015401 (2017). **DOI**: 10.1103/PhysRevMaterials.1.015401.
6. C. Nyshadham, C. Oses, J. E. Hansen, I. Takeuchi, S. Curtarolo & G. L. W. Hart, *A Computational High-Throughput Search for New Ternary Superalloys*, Acta Mater. **122**, 438–447 (2017). **DOI**: 10.1016/j.actamat.2016.09.017.
5. S. Sanvito, C. Oses, J. Xue, A. Tiwari, M. Žic, T. Archer, P. Tozman, M. Venkatesan, J. M. D. Coey & S. Curtarolo, *Accelerated Discovery of New Magnets in the Heusler Alloy Family*, Sci. Adv. **3**(4), e1602241 (2017). **DOI**: 10.1126/sciadv.1602241. [PDF]

2016

4. A. van Roekeghem, J. Carrete, C. Oses, S. Curtarolo & N. Mingo, *High-Throughput Computation of Thermal Conductivity of High-Temperature Solid Phases: The Case of Oxide and Fluoride Perovskites*, Phys. Rev. X **6**(4), 041061 (2016). **DOI**: 10.1103/PhysRevX.6.041061. [PDF]

3. K. Yang, C. Oses & S. Curtarolo, *Modeling Off-Stoichiometry Materials with a High-Throughput Ab-Initio Approach*, Chem. Mater. **28**(18), 6484–6492 (2016). **DOI**: 10.1021/acs.chemmater.6b01449.

2015

2. C. E. Calderon, J. J. Plata, C. Toher, C. Oses, O. Levy, M. Fornari, A. Natan, M. J. Mehl, G. L. W. Hart, M. Buongiorno Nardelli & S. Curtarolo, *The AFLOW Standard for High-Throughput Materials Science Calculations*, Comput. Mater. Sci. **108A**, 233–238 (2015). **DOI**: 10.1016/j.commatsci.2015.07.019.
   - This paper was selected for Editor's Choice by Elsevier (2015).
1. O. Isayev, D. Fourches, E. N. Muratov, C. Oses, K. M. Rasch, A. Tropsha & S. Curtarolo, *Materials Cartography: Representing and Mining Materials Space Using Structural and Electronic Fingerprints*, Chem. Mater. **27**(3), 735–743 (2015). **DOI**: 10.1021/cm503507h. [PDF]
   - This paper was one of the top 10 most highly downloaded papers for the month of January 2015 by the American Chemical Society (2015).
   - This paper was selected for Editors' Choice by the American Chemical Society (2015).

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Book Publications

2019

3. C. Toher, C. Oses & S. Curtarolo, *Automated computation of materials properties*, Materials Informatics: Methods, Tools and Applications, Ch. 7. **DOI**: 10.1002/9783527802265.ch7.

2018

2. S. Sanvito, M. Žic, J. Nelson, T. Archer, C. Oses & S. Curtarolo, *Machine learning and high-throughput approaches to magnetism*, Handbook of Materials Modeling. Volume 2 Applications: Current and Emerging Materials. **DOI**: 10.1007/978-3-319-50257-1_108-1.
1. C. Toher, C. Oses, D. Hicks, E. Gossett, F. Rose, P. Nath, D. Usanmaz, D. C. Ford, E. Perim, C. E. Calderon, J. J. Plata, Y. Lederer, M. Jahnátek, W. Setyawan, S. Wang, J. Xue, K. M. Rasch, R. V. Chepulskii, R. H. Taylor, G. Gomez, H. Shi, A. R. Supka, R. Al Rahal Al Orabi, P. Gopal, F. T. Cerasoli, L. Liyanage, H. Wang, I. Siloi, L. A. Agapito, C. Nyshadham, G. L. W. Hart, J. Carrete, F. Legrain, N. Mingo, E. Zurek, O. Isayev, A. Tropsha, S. Sanvito, R. M. Hanson, I. Takeuchi, M. J. Mehl, A. N. Kolmogorov, K. Yang, P. D'Amico, A. Calzolari, M. Costa, R. De Gennaro, M. Buongiorno Nardelli, M. Fornari, O. Levy & S. Curtarolo, *The AFLOW Fleet for Materials Discovery*, Handbook of Materials Modeling. Volume 1 Methods: Theory and Modeling. **DOI**: 10.1007/978-3-319-42913-7_63-2.

Software and Datasets

Prior development, Duke University (2014–2022)

aflow++ (2014–2022) · Lead developer · C++ framework for autonomous materials design and high-throughput ab initio workflows · with the AFLOW Consortium (S. Curtarolo, PI, Duke University) · **DOI**: 10.1016/j.commatsci.2022.111889 · aflow.org · Also developed the electronic band-gap and Bader charge analysis modules; POCC, CHULL and APL are listed separately below · Continuing development of a Johns Hopkins-specific branch (2022–present).

AFLOW-POCC (2014–2022) · Lead developer · partial-occupation and disorder module for modeling off-stoichiometric and high-entropy materials · with the AFLOW Consortium · **DOI**: 10.1021/acs.chemmater.6b01449 & 10.1016/j.actamat.2022.118594.

AFLOW-CHULL (2014–2022) · Lead developer · cloud-oriented platform for autonomous thermodynamic phase-stability analysis · with the AFLOW Consortium · **DOI**: 10.1021/acs.jcim.8b00393 · aflow.org/aflow-chull.

AFLOW-APL (2014–2019) · Core developer · automatic phonon library for vibrational and thermal properties · with the AFLOW Consortium · **DOI**: 10.1038/s41467-021-25979-5.

aflow.org (2014–2022) · Core developer · web ecosystem of materials databases, software and tools · with the AFLOW Consortium · **DOI**: 10.1016/j.commatsci.2022.111808 · aflow.org.

AFLOW-SYM (2014–2019) · Contributing developer · complete, automatic and self-consistent symmetry analysis of crystals · with the AFLOW Consortium · **DOI**: 10.1107/S2053273318003066 · aflow.org/aflow-online.

AFLOW-CCE (2021) · Contributing developer · automated coordination-corrected formation enthalpies · with the AFLOW Consortium · **DOI**: 10.1103/PhysRevMaterials.5.043803 · aflow.org/aflow-online.

AFLOW-ML (2018) · Contributing developer · RESTful API for machine-learning prediction of materials properties · with the AFLOW Consortium · **DOI**: 10.1016/j.commatsci.2018.03.075 · aflow.org/aflow-ml.

AFLUX (2017) · Contributing developer · search API for the AFLOW data repositories · with the AFLOW Consortium · **DOI**: 10.1016/j.commatsci.2017.04.036 · aflow.org/API/aflux.

OPTIMADE (2021–2024) · Contributor · community API standard for exchanging materials data across databases · with the OPTIMADE Consortium · **DOI**: 10.1038/s41597-021-00974-z & 10.1039/D4DD00039K · optimade.org.

Teaching Experience

Instructor — Spring 2023–present — EN.500.113: *Gateway Computing: Python*, Johns Hopkins University

- Required first-year coding course for all Whiting School undergraduates; one section in Spring 2023 and two sections every spring since, 20 to 32 students per year
- Project-based flipped classroom with in-class team problems solved at the whiteboard; authored one of the class-wide projects (string manipulation applied to order and disorder in materials)
- Course evaluations (seven sections, 91 respondents, response rates 86 to 100%): teaching effectiveness 4.25, intellectual challenge 4.34 of 5 (school means 4.27 and 4.28); Spring 2023 section 4.69 for teaching effectiveness

Instructor — Fall 2023–present — EN.510.466 / EN.510.666: *Materials Modeling (formerly Introduction to Computational Materials Modeling)*, Johns Hopkins University

- Cross-listed undergraduate and graduate course; enrollment 15 to 17 per year, mostly graduate students, with a waiting list in 2026
- Seven programming assignments with LaTeX reports in GitHub, no written examinations, final group project; closes with autonomous experimentation on the LEGOLAS kit
- Course evaluations (three years, 44 respondents, response rates 80 to 94%): quality 4.30, teaching effectiveness 4.30, intellectual challenge 4.62 of 5 (school mean for challenge about 4.26)

Research Advisor — Spring 2023–present — EN.510.501 / EN.510.502 / EN.510.809: *Undergraduate Research, Research in Materials Science, and Graduate Summer Research*, Johns Hopkins University

Guest Lecturer — Feb 05, 2026 — EN.510.603: *Phase Transformations (T. Curk)*, Johns Hopkins University

Guest Lecturer — Mar 28, 2023 — EN.510.422 / EN.510.622: *Micro and Nano Structured Materials and Devices (A. S. Hall)*, Johns Hopkins University

Co-Instructor — Spring 2021 — ME 555: *Applications of Artificial Intelligence in Materials*, Duke University Department of Mechanical Engineering and Materials Science

Teaching Assistant — Spring 2020 — ME 555: *Computational Materials Science by Examples and Applications*, Duke University Department of Mechanical Engineering and Materials Science

Teaching Assistant — Fall 2014–Spring 2015 — ME 221: *Structure and Properties of Solids*, Duke University Department of Mechanical Engineering and Materials Science

- Best Teaching Assistant Award, Aug 14, 2015

Conference and Workshop Organization † contributed equally

2026

30. *Laboratory to Launchpad: Bringing Your Energy IP to Life*. **Organizer** at Hopkins Bloomberg Center, Washington, DC (Jun 11, 2026). **Co-Organizers**: A. G. Bregman†, S. D. Cohen & J. Erlebacher.
    - website: https://engineering.jhu.edu/materials/laboratory-to-launchpad/
29. *Navigating Early Careers in Academia*. **Panelist** for the Materials Graduate Society at Johns Hopkins University, Baltimore, Maryland (Apr 15, 2026).
28. *Quantum Computing in Practice: Hands On with Quantinuum*. **Organizer** at Johns Hopkins University, Baltimore, Maryland (Feb 25–26, 2026). **Co-Organizer**: G. D. Quiroz.
    - website: https://entropy4energy.ai/workshops

2025

27. *Mini Symposium on "Computational Thermodynamics: Energy and Energy Landscape"*. **Co-Organizer and Presenter** at the 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, University Park, PA (Oct 31–Nov 2, 2025). **Co-Organizers**: Z.-K. Liu, J. Deng, T. R. Sinno & R. Wentzcovitch.
    - website: https://siamnnp2025.sciencesconf.org/